POISSON - REACTION

$$\begin{cases} -u'' + \sigma u = f & x \in [a,b] \\ u(a) = 0 \\ u(b) = 0 \end{cases}$$

$u \in H^1_0((a,b))$

$\Downarrow$

DIRICH. OMOGENEO

FORMA DEBOLE

$$\forall v \in H^1_0 \quad \int_a^b v \cdot (-u'' + \sigma u - f)\, dx = 0$$

$$\Downarrow$$

$$\int_a^b v' \cdot u'\, dx - \overbrace{\left[v \cdot u' \right]_a^b}^{=0} + \int_a^b \sigma v \cdot u = \int_a^b f \cdot v$$

$$v(a) = 0 = v(b)$$

$$\boxed{\int_a^b v' \cdot u' + \sigma v \cdot u \, dx = \int_a^b f \cdot v \, dx}$$

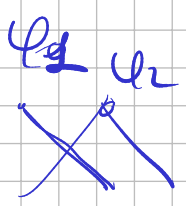
$$a(u,v) := \int_a^b v' u' + \sigma v \cdot u \, dx$$

$$F(v) := \int_a^b f \cdot v \, dx$$

ELEMENTI FINITI $\quad H^1_0 \to V_h := \left\{ v_h \in X^1_h : \begin{array}{l} v_h(a) = 0 \\ v_h(b) = 0 \end{array} \right\}$

$\forall v_h \in V_h$ TROVARE u_h

$$a(u_h, v_h) = F(v_h)$$



$$\to u_h(x) = \sum_{j=1}^{N_h} u_j \varphi_j(x)$$

$$\forall v_h \in V_h \iff \forall \varphi_i \in V_h \qquad \langle \varphi_i \rangle = V_h$$

$$i = 1, \dots, N_h$$

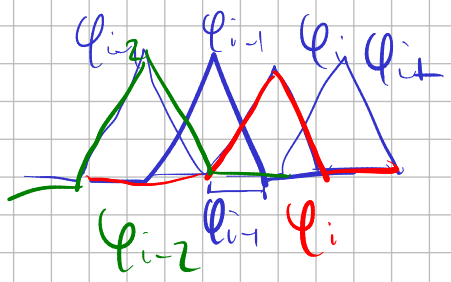
$$\overbrace{\sum_{j=1}^{N_h} \int_a^b u_j \cdot \varphi_j'(x)}^{u_h'(x)} \cdot \overbrace{\varphi_i'(x)}^{v_h'(x)} \, dx \qquad \forall i = 1, \dots, N_h$$

$$+ \sum_{j=1}^{N_h} \int_a^b \sigma \, u_j \cdot \varphi_j(x) \cdot \varphi_i(x) \, dx = \int_a^b f(x) \cdot \varphi_i(x) \, dx$$

$$\underline{\underline{A}} \cdot \underline{u} = \underline{f}$$

$$A = [a_{ij}] \qquad a_{ij} := \int_a^b \varphi_i'(x) \varphi_j'(x) \, dx + \int_a^b \sigma \varphi_i \varphi_j \, dx$$

$$f_i = \int_a^b f(x) \cdot \varphi_i(x) \, dx$$



$$a_{i-2,i} = 0$$

$$a_{i,j} \neq 0 \; ? \qquad j = \begin{cases} i-1 \\ i \\ i+1 \end{cases}$$

$$a_{i,i-1} = \int_a^b \varphi_i' \varphi_{i-1}' + \sigma \varphi_i \varphi_{i-1} \, dx =$$

$$= \int_{x_{i-1}}^{x_i} \varphi_i' \varphi_{i-1}' + \sigma \varphi_i \varphi_{i-1} \, dx \qquad \text{1 INTERVALLO}$$

$$a_{ii} = \int_a^b \varphi_i' \varphi_i' + \sigma \varphi_i \varphi_i \, dx = \underbrace{\int_{x_{i-1}}^{x_i} (\varphi_i')^2 + \sigma \varphi_i^2}_{} + \underbrace{\int_{x_i}^{x_{i+1}} (\varphi_i')^2 + \sigma \varphi_i^2}_{}$$

2 INTERVALLI

$$T_i : [0,1] \to K_i = [x_{i-1}, x_i]$$

$$x = T_i(\xi) = x_{i-1} + (x_i - x_{i-1}) \cdot \xi$$

$$\xi = T_i^{-1}(x) = \frac{x - x_{i-1}}{x_i - x_{i-1}} \qquad \frac{d\xi}{dx} = \frac{1}{x_i - x_{i-1}} = \frac{1}{h_i}$$

$$\frac{dx}{d\xi} = h_i$$

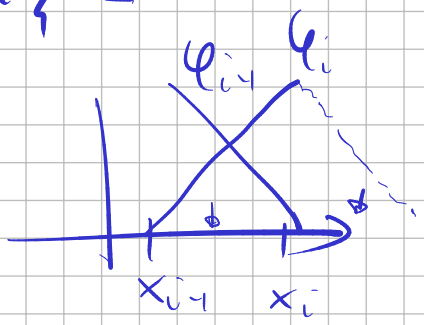
$$\boxed{dx = h_i\, d\xi} \qquad \frac{\partial}{\partial x} = \frac{\partial \xi}{\partial x} \cdot \frac{\partial}{\partial \xi}$$

$$\xi = T_i^{-1}(x)$$

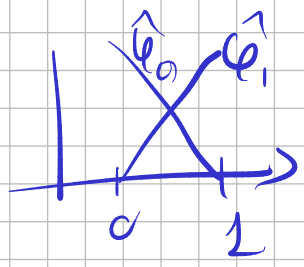
$$\int_{x_{i-1}}^{x_i} \partial_x \varphi_i(x) \cdot \partial_x \varphi_{i-1}(x) + \sigma\, \varphi_i(x)\varphi_{i-1}(x)\, dx$$

$$= \int_0^1 \left[\frac{\partial \xi}{\partial x} \cdot \partial_\xi \overbrace{\varphi_i(T_i(\xi))}^{\hat{\varphi}_1} \cdot \frac{\partial \xi}{\partial x}\, \partial_\xi \overbrace{\varphi_{i-1}(T_i(\xi))}^{\hat{\varphi}_0} + \sigma\, \varphi_i(T_i(\xi))\, \varphi_{i-1}(T_i(\xi)) \right] h_i\, d\xi =$$

$$= \int_0^1 \frac{1}{\cancel{h_i^2}} \partial_\xi \hat{\varphi}_1(\xi) \cdot \partial_\xi \hat{\varphi}_0(\xi)\, \cancel{h_i}\, d\xi + \int_0^1 \sigma\, \hat{\varphi}_1(\xi)\, \hat{\varphi}_0(\xi)\, h_i\, d\xi$$



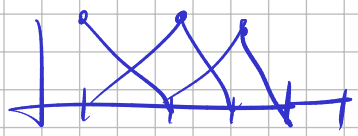
$$= \frac{1}{h_i} \int_0^1 \partial_\xi \hat{\varphi}_1(\xi) \cdot \partial_\xi \hat{\varphi}_0(\xi)\, d\xi + \underbrace{h_i}_{\text{VERSATILE}} \int_0^1 \sigma \cdot \hat{\varphi}_1(\xi) \cdot \hat{\varphi}_0(\xi)\, d\xi$$



CALCOLIAMO 1 VOLTA SOLA

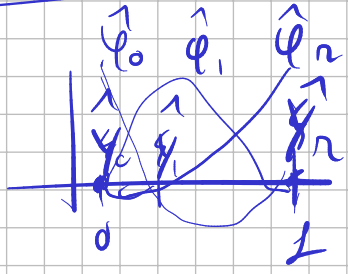
$a_{i,i-1} =$

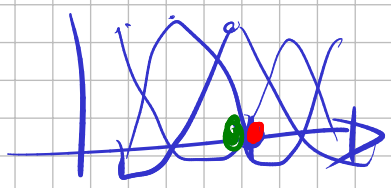
$a_{i,i} = a_{ii}^{[x_{i-1}, x_i]} + a_{ii}^{[x_i, x_{i+1}]}$



$a_{i,i+1}$

$[x_{i-1}, x_i] = K_i$





$y_\alpha = y(i,s)$

$= x_{i-1} + h_i \cdot \hat{y}_s$

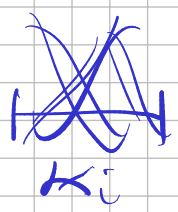
$N = \#$ elementi

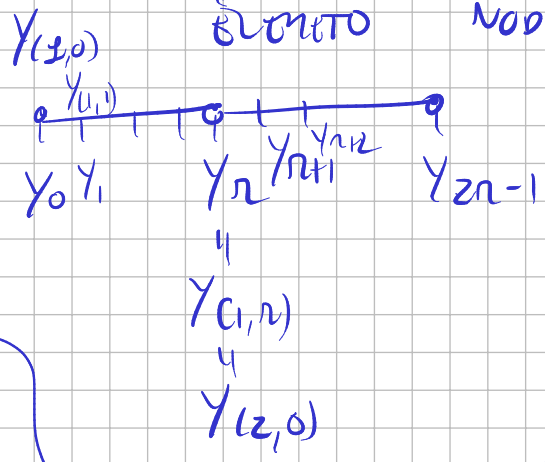
$s = 0, \dots, r$

$Y_{(i,0)} = Y_{(i-1,r)}$

$$\boxed{\alpha = (i-1)\cdot r + s}$$

ELEMENTO NODO.

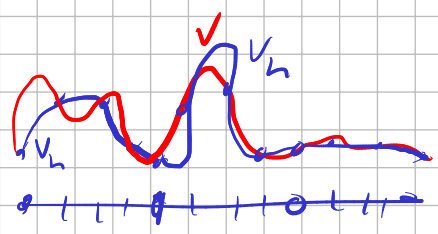




STIMA DELL'ERRORE

$v \in C^0((a,b))$ $\quad X_h^r \ni v_h$

$\Pi_h^r : C^0 \to X_h^r$



$v_h(x_i) = \Pi_h^r(v)(x_i) = v(x_i)$

$v_h = \Pi_h^r(v)$

TEOREMA $v \in H^{r+1}((a,b))$ CON $r \geq 1$

($H^1 \subset C^0$ QUINDI HA SENSO PARLARE DI INTERPOLAZ)

$$|v - \Pi_h^r v|_{H^0((a,b))} \leq C_{0,r}\, h^{r+1} \cdot |v|_{H^{r+1}((a,b))}$$

$$|v - \Pi_h^r v|_{H^1((a,b))} \leq C_{1,r} \cdot h^r\, |v|_{H^{r+1}((a,b))}$$

$C_{0,r}$ e $C_{1,r}$ INDIPENDENTI DA h e da v

$X^1 \Rightarrow \| \ \|_{L^2} \approx h^2$ $\qquad | \ |_{H^1} \approx O(h)$

STIMA DELL'ERRORE

$u \in V \subset H^1((a,b))$ SOL EX. PROBLEMA DEBOLE

e $u_h \in V_h = V \cap X_h^r$ SOL ELEMENTI FINITI

$$\Rightarrow \|u-u_h\|_{H^1} \leq \frac{M}{\alpha} C h^r |u|_{H^{r+1}((a,b))}$$

C dip da u, h

M CONTIN. DI FORMA BILINEARE a

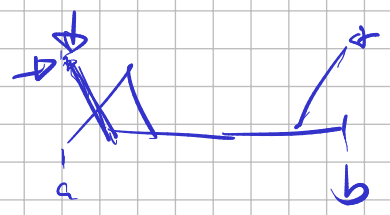
α COERCIVITÀ DI " " a

DIM LEMMA DI CÉA

$$\|u-u_h\|_V \leq \frac{M}{\alpha} \cdot \inf_{v_h \in V_h} \|u-v_h\|_V \leq \frac{M}{\alpha} C_{1,r} h^r |u|_{H^{r+1}((a,b))}$$

$\boxtimes$

CONDIZIONI AL BORDO



DIRICHLET $u(a) = u_L$, $u(b) = u_R$

$A \leftarrow$ TUTTE LE FUNZIONI DI BASE (INCLUSE BORDO)

$$\begin{cases} a_{11} = 1 & a_{1,j} = 0 \quad \forall j > 1 \\ f_1 = u_L \end{cases}$$

DIRICHLET A SX

DX $a_{N_h N_h} = 1$ $a_{N_h, j} = 0$ $\forall j < N_h$

$f_{N_h} = u_R$

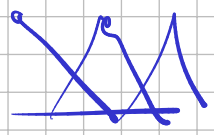
NEUMANN $\quad u'(a) = g \qquad u(b) = \beta \qquad \varphi \in V = H^1 : \underline{\underline{\varphi(b) = 0}}$

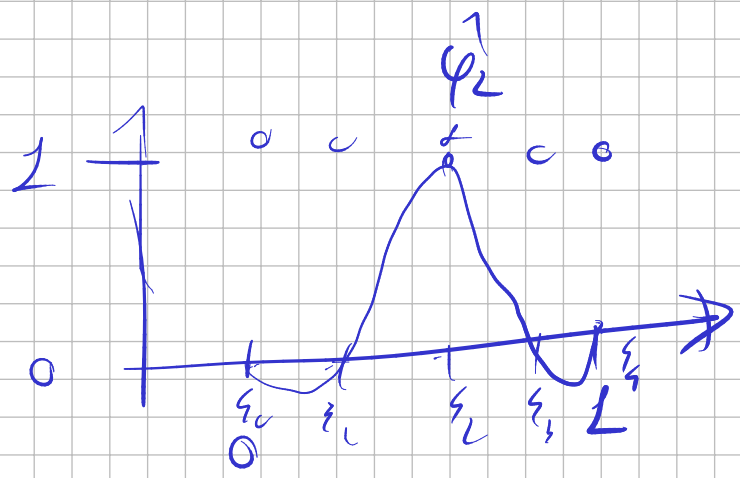
$$\int_a^b f \varphi_i = \underbrace{\int_a^b -u'' \cdot \varphi_i \, dx} = \int_a^b u' \cdot \varphi_i' \, dx - \left[u' \cdot \varphi_i \right]_a^b$$

$$= \int_a^b u' \varphi_i' \, dx - \left(u'(b) \underbrace{\varphi_i(b)}_{=0} - \underbrace{u'(a)}_{g} \cdot \varphi_i(a) \right)$$

$$= \underbrace{\int_a^b u' \varphi_i' \, dx}_{a} + \underbrace{g \cdot \varphi_i(a)}_{RHS}$$

$$F(\varphi) = \int_a^b f \cdot \varphi - g \, \varphi(a)$$



$$\varphi_i(a) = \begin{cases} 0 & i > 0 \\ 1 & i = 0 \end{cases}$$

$$f_i = F(\varphi_i) = \begin{cases} \int_a^b f \cdot \varphi_i & i > 0 \\ \int_a^b f \cdot \varphi_0 - g \underbrace{\varphi_0(a)}_{=1} & i = 0 \end{cases}$$